

DEFYING SCALING IN POTENTIAL THEORY

IWONA CHLEBICKA
UNIVERSITY OF WARSAW

Classical non-linear potential theory thrives under the predictable comfort of power-law scaling, where p -Laplacian operators yield well-behaved pointwise estimates through traditional Wolff potentials. But what happens when homogeneity breaks down completely? In this talk, we defy the constraints of standard scaling to tackle measure data elliptic problems governed by second-order operators with non-standard Orlicz growth and merely measurable coefficients.

We break away from homogeneous paradigms by establishing sharp pointwise estimates expressed in terms of novel, generalized potentials tailored to non-scaling structures. Going beyond describing local solution behavior where standard tools fail, we uncover the exact regularity consequences dictated by these estimates. Crucially, this framework holds across both scalar equations and vectorial systems, forging a powerful potential theory liberated from scaling laws.